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The Nature and Culture of Physical Activity Play: Towards a Biocultural Model of Play, Practice, and Teaching in Sports

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Play behavior continues to fascinate researchers from various fields. This has led to several domain-specific bodies of literature with limited cross-disciplinary integration. The goal of the present review was to integrate evolutionary literature on physical activity play with literature from sport science and pedagogy as these fields can mutually benefit from one another. The resulting literature synthesis showed that the evolutionary theorizing on the functions of physical activity play (physical and social skill training, training-for-the-unexpected, and fostering domain-specific creativity and innovation) proved helpful in explaining how diversified play experiences at an early age contribute to positive development and success in sports. Further, the review of sports data provided unique support for the developmental scaffolding theory that the innate motivation of seeking pleasure in physical activity play coincidentally leads to the acquisition of physical skills and enhanced domain-specific creativity. The cross-disciplinary literature review resulted in a novel biocultural model of skill acquisition and teaching with practical applications in sports and physical education. Potential avenues for further cross-disciplinary basic and applied research are discussed.

Public Significance Statement

The prevalence of physical activity play in animals and humans suggests that the biological benefits of this facet of behavior outweigh the costs. The reviewed research suggests that the innate predisposition to enjoy engaging in physical activity play behavior at early ages leads to beneficial outcomes in nature and in the human performance domain of sports. Early engagement in physical activity play benefits animals and athletes by training physical skills and producing novel behaviors that allow individuals to cope with a variety of unexpected circumstances. Therefore, sports programs should exploit play activities as a natural means of learning perceptual-motor skills and adapting to unpredictable environments. Early diversified play experiences should be encouraged, whereas early specialized practice activities, and explicit, verbal teaching of perceptual-motor skills at early ages should be deemphasized.

Keywords: sport, evolution, play, physical activity, culture

> Nobody has to teach young mammals to play. They come into the world designed for it (Gray, 2018, p. 84).

Play can be considered important from an evolutionary perspective as it occurs in a variety of species (Burghardt, 2005; Burghardt & Pellis, 2018; Graham & Burghardt, 2010; Pellegrini et al., 2007; Pellis et al., 2015; Sutton-Smith, 1997). Surprisingly this evolutionary literature on play is fairly neglected in the applied field of sports. Sports data has also hardly been used to test evolutionary theorizing on play. In the present review I argue that both fields can mutually benefit from one another. An evolutionary perspective on play might be helpful in explaining emerging evidence in the sports literature showing that diversified play experiences at an early age are correlated with

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positive development and success in sports (e.g., Güllich, 2014, 2017; Güllich & Emrich, 2014; Hornig et al., 2016). On the other hand, sports data might provide unique support for functional hypotheses of play found in the evolutionary literature. Therefore, the present article aims to integrate evolutionary literature on physical activity play (Pellegrini & Smith, 1998) with literature from sports science. The goal is to provide a biocultural model of skill acquisition and teaching that has practical application in sports and physical education.

The Evolution of Physical Activity Play

Definitions of play often start to define play by the inability to satisfactorily define it (Lewis, 2010, p. 61). The five categorical criteria put forth by Burghardt (2005, 2010) for recognizing play in all species (including humans) are a helpful working definition of play: (a) play is incompletely functional (i.e., does not serve a specific purpose) in the context it occurs; (b) play is spontaneous, enjoyable, voluntary, and rewarding as a means, not an end. This defining feature of play is nicely captured by the quote of Huizinga (1955, p. 3) that "fun characterizes the essence of play"; (c) play can be distinguished from "more serious" behaviors like hunting or fighting (e.g., it is exaggerated; it coincides with distinct nonverbal expressions) and in timing (e.g., occurs more frequently early in development); (d) play is repetitive, but not pathological like other repetitive behaviors (e.g., rocking or pacing); (e) play is initiated in the absence of stress or immediate threat. For a behavior to be classified as play as opposed to another form of behavior, all five criteria have to be met (Pellegrini, 2009; Pellis & Pellis, 2009).

The defining feature of physical activity play is that it involves gross motor behavior in a playful context with a metabolic rate above the resting metabolic rate (Pellegirini & Smith, 1998). Examples of physical activity play include chasing, play wrestling, climbing, and throwing and catching objects. A widely cited definition that captures the essence of physical activity play was proposed by Bekoff and Byers (1981):

> Play is locomotor activity performed postnatally which appears to an observer to have no obvious immediate benefits for the player, in which motor patterns resembling those used in serious functional contexts may be used in modified terms. The motor acts constituting play have some or all of the following structural features: exaggeration of movements, repetition of motor acts, and fragmentation or disordering of sequences of motor acts. (p. 301)

Potential Evolutionary Functions of Physical Activity Play

Why would natural selection have promoted a class of behaviors in a variety of species (Burghardt, 1998, 2005; Graham & Burghardt, 2010; Pellegrini et al., 2007; Sands & Sands, 2010) that, by definition, looks purposeless and clearly has biological costs (Auerbach et al., 2015; Yanagi & Berman, 2018)? Costs of play include excessive energy expenditure (Miller & Byers, 1991), time lost from productive activities like foraging (Barber, 1991), and a greater exposure to risks such as injury and predation (Harcourt, 1991). The systematic approach of providing evidence for evolutionary functions of behavior was put forth by Tinbergen (1952, 1963) who proposed four seminal questions that can be considered an important tool to understand any product of evolution: First, what are the physical mechanisms leading to play behavior, or what causes individuals to play? Second, how does play behavior develop during the lifetime of the organism? Third, what is the history of play as it evolved over multiple generations? Fourth, what is/are the function(s) of play, or stated differently, how does playing allow humans to adapt to their environment, and how does playing influence reproductive fitness, or output?

There are likely major differences between species in the functions, mechanisms, and development of play (Pellegrini et al., 2007; Sands & Sands, 2010; Smith & Roopnarine, 2018). Trezza et al. (2018) have for example detailed the neurophysiological mechanism in rats (first question) that orchestrate the adaptive motivation to engage in play. These mechanisms include cascades of neurotransmitter releases (i.e., opioids, endocannabinoids, dopamine, serotonin, and noradrenaline) that are regulated through a distributed network of brain regions in the frontal cortex and limbic system. Neuroendocrinological research has shown that play is directly associated with the release of opioids in the brain making it an inherently rewarding and pleasurable experience (Trezza et al., 2010). For this reason, the animals are motivated to spend substantial amounts of time with play behavior.

Regarding the second question, research has suggested that play behavior tends to be more prevalent at early developmental stages (Smith & Roopnarine, 2018 for a review). However, this developmental trajectory varies across species and play activities. The higher prevalence of play behavior in early periods of development have led to the theory of developmental scaffolding (Bateson, 1976). As a byproduct of enjoying play and engaging in play, individuals are assumed to have the opportunity of acquiring and practicing important skills (Gray, 2018; Groos, 1898, 1901). Once certain skills have been learned and mastered, play activities are reduced or fall away completely. This scaffolding theory is supported by evidence showing an inverted-U developmental course of play behavior (second question) that begins in early infancy, peaks during childhood, and then declines in adolescence and almost disappears in adulthood (Byers & Walker, 1995; Fagen, 1981; Rubin et al., 1983). Given its enjoyable nature and rewarding properties, likely immediate benefits of play are positive emotions and pleasure evoked by play (see first question). In the long run, this may facilitate the development of brain networks that are used during the playful activity: for example, networks underlying motor control and coordination (Berghänel et al., 2015; Byers & Walker, 1995), social functioning (Pellis & Pellis, 2009), affect and emotion (Trezza et al., 2018; Vanderschuren & Trezza, 2014), and creativity (Bateson, 2015). This reasoning is based on evidence that play occurs more frequently in periods (second question) in which neuronal development may be modified (Bjorklund & Green, 1992; Bjorklund & Pellegrini, 2002; Byers & Walker, 1995; Gomendio, 1988; Gray, 2018; Pellegrini et al., 1998; Sharpe, 2018; Thelen, 1979). Pertinent to this reasoning, the nervous system of mammals shows a superabundance of connections between neurons at the start of development. Neural connections that are frequently used in the experiential activities the organism engages in are retained, and the unused ones are lost. This process has been summarized as *proliferate and prune* (e.g., Sapolsky, 2017 for a review) which has, for example, received empirical support by studies on rats (Bell et al., 2010) and hamsters (Burleson et al., 2016). These studies showed morphological changes in the brains of the animals as a function of social play that were accompanied by functional behavioral consequences.

Although the third question has not received the same amount of research attention as the other questions, there is evidence to suggest that a high prevalence of play behavior tends to coincide with an extended period of immaturity in species (Bateson, 2005; Gray, 2018; Fagen, 1981; Lancaster & Lancaster, 1987; Pellis et al., 2014; Sutton-Smith, 1997). Further, there is evidence for a correlation showing that the more complex and flexible an organism behaves, the longer the period of immaturity is (Bjorklund, 2006; Bjorklund & Rosenberg, 2005). Taken together, this evidence has led to the assumption that play in immaturity is an important evolutionary strategy to develop behaviors that are adaptive to the niches the species inhabit. If these niches require large degrees of behavioral flexibility, then play behavior tends to be more prevalent in the species (Burghardt, 1998; Fagen, 1981).

How play improves animals' chances of passing on their genes to the next generations is a debated topic (Gray, 2018; Sharpe, 2018 for recent reviews). Play behavior has been suggested to have several immediate (second question) and delayed functions (or ultimate functions; fourth question; Cameron et al., 2008; Fangen & Fangen, 2009; Nunes, 2014; Pellis & Pellis, 2009; Vanderschuren & Trezza, 2014). For example, studies on play deprivation show an increased vulnerability to the effects of social stress and thereby arguably impact reproductive fitness (e.g., Burleson et al., 2016). The next sections will discuss more specific hypotheses on the functional importance of physical activity play. As numerous functions have been proposed and play may serve different functions in different species (Sharpe, 2018; Smith & Roopnarine, 2018), I will focus on four functions proposed in the evolutionary literature on physical activity play that arguably are important for positive development and success in sports: (a) practice of physical skills; (b) practice of social skills; (c) training for the unexpected (both physically and emotionally); and (d) fostering creativity and novel behaviors.

Practice of Physical Skills

One of the most trivial assumed beneficial consequences of physical activity play is the training of physical skills. Somewhat different to the typical inverted-U developmental course of general play behavior, physical activity play in humans shows three successive peaks with different types of physical play that probably serve different developmental functions: (a) rhythmic stereotypies; (b) exercise play; (c) Rough-and-Tumble play (R & T; Pellegrini & Smith, 1998); and (d) coalitional play fighting.

Rhythmic Stereotypies

Examples of rhythmic stereotypies include arm waving, body rocking, or foot kicking. Although it remains controversial whether these behaviors can be classified as play, they are similar to the defining features of physical activity play as both involve gross body movements, with above resting metabolic rate, and it is difficult to ascribe goal or purpose to these movements (Thelen, 1979). Rhythmic stereotypies are most likely controlled by neuromuscular maturation and peak in the first six months of infant development and gradually disappear thereafter (Thelen, 1980). The rhythmic movements of certain body parts increase before infants achieve voluntary control over these body parts, suggesting that these movements are beneficial for improving motor control (Pellegrini & Smith, 1998). In this respect, these behaviors seem to play an important role in the *proliferate-and-prune* process (e.g., Sapolsky, 2017 for a review) in which the central nervous system initially overproduces synapses (largely gene-driven) which is followed by a pruning back of the unused and overabundance of synapses (largely experience-driven; Byers & Walker, 1995; Thelen, 1979).

Exercise Play

Exercise play occurs later than rhythmic stereotypies and can be described as physically vigorous movements in a playful context that are either solitary or social. This type of play often starts at the end of the first year after locomotion has been achieved (i.e., walking in humans), increases during the preschool years, and declines during primary school (Eaton & Yu, 1989; Routh et al., 1974). Males engage in more exercise play than females, with the effect size increasing from infancy to the middle of adolescence (Eaton & Enns, 1986; Pellegrini & Smith, 1998). Exercise play is assumed to benefit muscle differentiation, strength, and endurance (Brownlee, 1954; Gray, 2018; Pellegrini & Smith, 1998; Sharpe, 2018). There is also evidence, suggesting that exercise play can serve beneficial functions for muscle coordination, efficiency, and, given sufficient frequency for strength and endurance (see Pellegrini & Smith, 2005; Sharpe, 2018 for reviews).

Rough-and-Tumble Play

R & T can be regarded as a social subcategory of exercise play. The term rough and tumble play was coined to describe play chasing, fleeing, and wrestling in rhesus monkeys (Harlow & Harlow, 1965), but has been extended as this type of play occurs in many other animal species (Aldis, 1975; Einon & Potegal, 1991; Fagen, 1981; Pellis et al., 1996; Smith, 1982; Smith & Roopnarine, 2018). In general, it is characterized by vigorous activities, by the absence or infrequency of threats, by free and easy movements, a relaxed muscle tone, inhibition of biting and other forms of violence, play signals such as the play face and play vocalizations, a frequent reversing of roles, and relaxed dominance relations (Aldis, 1975; Bekoff & Byers, 1981; Fagen, 1981; Smith, 1982; Symons, 1978). The developmental path of R & T is similar to exercise play in humans starting at the end of the first year, increasing during preschool, and declining during primary school (Pellegrini & Smith, 1998).

There are substantial gender differences in both exercise play and R & T (Pellegrini & Smith, 1998). No gender differences are apparent in rhythmic stereotypies (Thelen, 1980), whereas they are significant in exercise play and R & T (Eaton & Enns, 1986). As fighting and hunting are considered activities that were typically carried out by men (Boulton & Smith, 1992) it is assumed that R & T serves the function of acquiring and refining fighting and hunting skills as these were typically male activities in the evolutionary history of humans (Smith, 1982; Symons, 1978). Evidence for this hypothesis comes from a study with wild macaques (Berghänel et al., 2015). This study showed that infants that spent more time in vigorous R & T displayed certain climbing, running, and leaping skills at a younger age than infants that engaged in less R & T.

Coalitional Play Fighting

Although dyadic play fighting (i.e., one-on-one) occurs in many species, it has been suggested that only humans engage in coalitional play fighting (Scalise Sugiyama et al., 2018). Coalitional play fighting is defined as playful activity in which coalitions compete against each other by using coordinated action to achieve a goal and preventing the opposition from reaching their goal. It is assumed that coalitional play fighting emerged as a consequence of intergroup aggression among human ancestors. Several lines of evidence suggest that warfare, largely in the form of raiding, may have been a prevalent feature in the life of early humans (reviewed in Scalise Sugiyama et al., 2018). Raiding has been defined as group of allied men cooperatively invading an outgroup territory in a strategic manner (Wrangham, 1999). Research has shown that coalitional play fighting evolved in humans across a diverse range of hunter-gatherer cultures and habitats as a means of practicing and refining motor skills involved in raiding, fighting, and attacking as a precursor of human warfare (Scalise Sugiyama et al., 2018). In addition, it seems likely that coalitional play fighting also constitutes a means of practicing and refining social skills such as cooperation, seeking out vulnerable opponents, and continually reassessing comparative fighting formidability of allies and opponents. According to Scalise Sugiyama et al. (2018) humans have therefore evolved an adaptive motivation to engage in coalitional play fighting to practice these skills. This reasoning is supported by gender differences in the development and maintenance of competitive coalitions (i.e., more pronounced in boys and men; Geary et al., 2003), by a greater prevalence of male coalitional play fighting (Scalise Sugiyama et al., 2018), and a greater popularity of watching and engaging in team sports (Furley, 2019).

Practice of Social Skills

Beside the hypothesized social functions of coalitional play fighting, R & T has been proposed to have functional significance on social competence (Pellegrini & Smith, 1998, 2005). R & T is assumed to be distinct from actual aggression and is not usually correlated with real aggressive behavior (Blurton Jones, 1972; Fry, 1987, 1992, 2005; Pellegrini, 1988). As unrestrained physical fights pose the risk of physical harm, R & T could be regarded as an activity to test and display desirable qualities like strength, endurance, bravery, and fighting ability (see Lombardo, 2012 for a similar line of reasoning). In this respect, children (especially boys) can display their qualities and gain in status and dominance by competing in R & T without actual physical harm. Therefore, R & T has been proposed to fulfill dominance functions in humans (Pellegrini & Smith, 1998). This reasoning is supported by the observation that R & T is most prevalent in the middle-childhood period, in which children are highly concerned about establishing status in peer groups (Waters & Sroufe, 1983). Certain animal species have also been shown to gain in dominance status by engaging in R & T (Blumstein et al., 2013). R & T as a means of acquiring dominance and status is further supported by substantial gender differences. Boys are assumed to gain dominance and status in social groups via physical prowess (Benenson, 2013 for a review; Pellegrini, 1995), whereas girls tend to primarily use verbal rather than physical means in this endeavor (Charlesworth & Dzur, 1987). By engaging in R & T boys are assumed to develop a better understanding of social competition and acquiring means for navigating hierarchies (Sharpe, 2018).

A further social skill among humans that might benefit from R & T is the ability to communicate nonverbally. Therefore, it seems feasible that R & T might contribute to developing and practicing this skill, for example, by encoding and decoding nonverbal cues that signal "this is play" and not real aggression (Bateson, 1972; Bekoff, 1995). Social signals, such as smiling and laughing frequently occur in the context of R & T and have been shown to function as such signals and thereby help to facilitate social group life and maintain cohesion in groups (Pellegrini & Smith, 2005).

Moreover, Gray (2018) reviewed evidence indicating that some forms of physical activity play may function to reduce hostility and foster cooperation in groups. By engaging in social forms of physical activity play, animals and humans can practice to bond, form coalitions, and cooperate to achieve a common goal. Tentative support for such reasoning comes from studies showing that both juvenile and adult pack-hunting animals frequently engage in social play (Cordoni, 2009). Further studies with macaques suggested that social play helped individuals to adapt to prevailing group structures by practicing to coexist peacefully in close contact (Reinhart et al., 2010) and fitting in with prevailing dominance hierarchies (Ciani et al., 2012). Studies with Bonobos (Palagi, 2006, 2008; Palagi et al., 2006; Palagi & Paoli, 2007) also suggested that social play facilitated the capacity to cooperate and reduced the risk of conflict in group life. Anthropological analyses have also found correlations in hunter-gatherer societies between measures of cooperation and frequency of social play (Gray, 2018).

Training for the Unexpected

Physical activity play (and other forms of play) have been suggested to enable animals to develop flexible kinematic and emotional responses to unexpected events (Spinka et al., 2001). Animals can practice a temporary loss of control and practice behaviors to regain control in a variety of situations. The training-for-the-unexpected hypothesis assumes that physical activity play increases the versatility of movements that an individual can use to recover from shocks during fight or flight activities (e.g., falling when being chased). Beside this motor versatility argument, individuals can also learn to deal with the emotional aspects of being surprised, disoriented, or temporally disabled through physical activity play (Spinka et al., 2001).

Although there is not much direct evidence for the training-for-the-unexpected hypothesis (Sharpe, 2018), two studies with squirrels (Marks et al., 2017; Nunes et al., 2004) showed that forms of social physical activity play were associated with a better ability to handle novel and unfamiliar situations. Further indirect evidence was reviewed by Gray (2018) who suggested that the frequency of dangerous play behavior (e.g., Pellis & Pellis, 2011) indicated that individuals frequently put themselves into vulnerable positions to practice getting out of these or coping emotionally with these situations. In this respect, Sandseter and Kennair (2011) identified six categories of risky play behavior (heights, high speed, dangerous tools, dangerous elements, rough-and tumble, disappearing/getting lost) that appeared to be universal among children, lending support to the idea of an evolutionary explanation for risky play as a means of acquiring physical and emotional skills to cope with the unexpected. Animal models have supported such reasoning by showing that individuals who engaged in more R & T showed a reduced cortisol response (a sign of less distress) in response to stressful situations (Mustoe et al., 2014). In addition, research showed that rats deprived of social play were unable to deal effectively with social stressors (Baarendse et al., 2013), had higher levels of stress hormones (von Frijtag et al., 2002), and showed greater anxiety in unfamiliar environments (Arakawa, 2003). Research in humans has also reported a correlation between the quantity childhood of play experience and the ability to cope with stressful events (Saunders et al., 1999).

Fostering Creativity and Innovation

Play has further been suggested to foster creativity and innovation (Bateson, 2015; Gray, 2018). According to this view, an evolutionary function of play could be to generate novel behaviors, some of which turn out to be useful in terms of survival and reproductive success. Bateson and Martin (2013) reviewed a large body of evidence on humans and animals that has suggested a link between engaging in different forms of play and producing creative solutions in a variety of domains. Their conclusion from this extensive literature review was that playful behavior can generate novel approaches to challenges set by the physical and social environment.

An early study (Birch, 1945) showed that a chimpanzee who previously had the opportunity to play with sticks, learned how to use these sticks to solve problems in the future. By playing with the sticks earlier, chimpanzees figured out how these sticks could be joined together to reach a desired banana that was otherwise out of reach. Chimpanzees that did not have the opportunity to play with sticks did not figure out how to reach the banana. Further studies with macaques indicated that playing with stones led to the generation of novel behaviors that are not only useful to the individual but are also subsequently passed on to peers and future generations (Huffman et al., 2008). Several studies have shown that playful interaction with objects can produce useful new behaviors in acquiring food. For example, bottlenose dolphins have been observed to initially play with empty shells of giant snails and subsequently use these shells to catch prey (Morell, 2020). Subsequently this new creative behavior is passed on to their peers and becomes prevalent in the group of bottlenose dolphins. A further study with Macaques (Tan, 2017) with the compelling title "from play to proficiency" has shown that initially playing with stones led to subsequent skillful behavior of cracking open shellfish.

Physical Activity Play in the Development of Skill in Sports

The reviewed evolutionary functions (practicing physical and social skills, training for the unexpected, and fostering creativity and novel behavior) show considerable overlap with training and developmental goals that have been proposed in the sports literature (e.g., Côté et al., 2007; Côté & Vierimaa, 2014; Memmert et al., 2015). As human culture should harness what individuals are born with (Boyd & Richerson, 1985; Richerson & Boyd, 2005), physical activity play seems to offer a useful means for learning these desired skills. In the next section I will review evidence on how physical activity play could help to develop important sports skills that might not be easily acquired with certain forms of sport-specific practice.

The Development of Skill and Success in Sports

A major topic of interest is how people achieve high levels of skills or even expertise in fields like sports. This topic is deeply embedded in the long-standing nature versus nurture debate (e.g., Ridley, 2003) that concerns the relative influence of innate factors versus learning and experience in determining skill level or expertise.

Deliberate Practice

The dominant view in the sports literature has been that the most important factor for acquiring high levels of sports skills is domain-specific practice: "Practice is uniformly regarded in the motor learning literature as the variable having the greatest singular influence on skill acquisition . . ." (Côté et al., 2007, p. 184). This view has been inspired by the influential article from Ericsson et al. (1993) on deliberate practice. According to the authors, the single "ingredient" for becoming an expert in any field is to accumulate a sufficient amount of deliberate practice: "The individualized training activities specifically designed by a coach or teacher to improve specific aspects of an individual's performance through repetition and successive refinement" (Ericsson & Lehmann, 1996, pp. 278–279). This deliberate practice view has been influential in the popular press and has been covered in several best-selling books including Gladwell's (2008) *Outliers* or Colvin's (2008) *Talent Is Overrated.* Such books have led many to accept what Gladwell termed the "10,000-hour rule"—the idea that it takes 10,000 hr of practice to become an expert in for example, sports.

Research shows that parents, athletes, and coaches believe that early single-sport specialization with large amounts of deliberate practice is necessary for success leading to a focus on short-term results at early ages rather than the overall development process (see DiFiori et al., 2018; Güllich & Emrich, 2014; Rees et al., 2016 for reviews). However, empirical research in athletes does not provide much support for these impressions. A first line of research has shown that although coach-led deliberate practice at young ages is associated with competitive success at the youth level, it is hardly correlated with long-term senior success (Barreiros et al., 2014; Brouwers et al., 2012; Güllich & Emrich, 2014; Vaeyens et al., 2009). Short-term youth success is correlated with early single sport specialization and deliberate practice (DiFiori et al., 2018; Ford et al., 2009; Güllich & Emrich, 2014). However, long-term success seems to be not dependent on early specialization and the accumulation of large amounts of deliberate practice. For example, world class athletes have been shown to accumulate relatively less sport-specific deliberate practice during childhood (Güllich, 2017; Moesch et al., 2011), but typically participated in relatively more diverse sport activities, including peer-led play and practice in a variety of sports (see Güllich & Emrich, 2014 for an over-view of this research). World-class athletes were more likely than national-class peers to play multiple sports (Güllich, 2017; Rees et al., 2016) and specialized in their primary sport significantly later than their national-class peers (Güllich, 2014, 2017; Güllich & Emrich, 2006). These findings were consistent across different countries and types of sports (Güllich & Emrich, 2014) and have even been corroborated when comparing Olympic and World Championship medalists to nonmedalists (Güllich, 2017; Rees et al., 2016). Although several factors might contribute to the data pattern of these studies (including e.g., innate talent or genetics), they do suggest that training activities and participation patterns that likely lead to short-term success early on are not the same as those that facilitate long-term development and adult success (DiFiori et al., 2018; Güllich & Emrich, 2014).

Of particular relevance in highlighting the importance of play in developing sport expertise is a study investigating the "microstructure" of practice of German world-class soccer players as it highlights the significance of play in achieving sport expertise (Hornig et al., 2016). The study showed that only 14% of their total childhood soccer activities involved deliberate practice activities (drill-like training of technical skills or physical conditioning) and as much as 86% was a combination of coach-led play (17%; including small-sided games), and peer-led play (69%; "kicking around with friends"). This finding is in line with a growing body of evidence demonstrating the importance of play (as opposed to practice) activities in acquiring a range of skills and even in achieving higher levels of sports expertise (Côté, 1999; Côté et al., 2007; Haugaasen et al., 2014; Memmert et al., 2010). Further studies suggest that playing multiple sports before specializing in a single sport is positively associated with positive development and future sport performance (Baker, 2003; Berry et al., 2008; Fransen et al., 2012; Güllich, 2014; Soberlak & Côté, 2003; but see Ford et al., 2009; Ford & Williams, 2012).

Meta-analytic evidence has also challenged deliberate practice theory in sports as deliberate practice only accounted for 18% of variance in sports skill (Macnamara et al., 2016) and only 1% of variance in performance among elite athletes. Hence, there is converging evidence that deliberate practice is only one factor in acquiring sport expertise (Hambrick et al., 2016) and physical activity play at an early age is also positively associated with the development of sport skill.

Deliberate Play

In recognition that an athlete's first sport experience is typically characterized through fun and play, Côté (1999) coined the term deliberate play. This term includes all forms of sporting activities that are intrinsically motivating, provide immediate gratification, and are designed by coaches and trainers to maximize enjoyment. Typically, deliberate play activities are governed by rules that are adapted from standardized sport rules. Côté (1999) distinguishes deliberate play from other forms of play like the pedagogical games designed to improve sport performance (Griffin & Butler, 2005), and the structured and "deliberate" practice activities typical of organized sport (Baker & Young, 2014). According to Côté et al. (2007) deliberate play shares the contextual characteristics of physical activity play (such as running, wrestling, R & T), but occurs in a more organized setting and typically entails more unique behavioral patterns (e.g., ball throwing).

Similar to the definitional problems encountered when discussing physical activity play, the term deliberate play is used loosely in the sports literature. The term emerged due to the prominent concept of deliberate practice (Ericsson et al., 1993), but can be considered somewhat misleading regarding defining features of play as being incompletely functional, voluntary, and spontaneous (Burghardt, 2005, 2010). These defining features seem contradictory to the term deliberate. Côté et al. (2007) also seem to have lost some of their original features of deliberate play (e.g., being designed by a coach or adult) by using the example of a child transitioning from playing alone with a ball at a backyard basketball hoop (i.e., not designed by a coach or adult), to shootout contests with friends, to one-on-one and two-on-two street basketball games. Hence, it seems to be a fair assessment to state that the term deliberate play is missing precise boundary conditions and mainly functions as a construct to distinguish between sport-specific play and practice activities. I will therefore continue to use the term physical activity play when discussing play activities in sport.

Why Is Play Associated With Positive Development and Success in Sport?

In addressing this question, I return to the four functional hypotheses outlined in the evolutionary literature review: practicing physical and social skills, training-for-the-unexpected, and fostering creativity and innovation. Research in the sports domain has provided evidence that physical activity play might contribute to sports expertise by implicitly practicing physical skills and by enhancing domain-specific creativity. It seems feasible to hypothesize that physical activity play in early years would also train social skills (e.g., getting along better with teammates, facilitated transition when changing teams, navigating hierarchies in teams, choosing vulnerable opponents to go one-on-one in team-sports, etc.) and prepare players for the unexpected (e.g., physically and emotionally cope with unpredictable situations emerging during play or their careers). These skills might arguably contribute to success later in an athlete's career. However, I am not aware of any research in sports that can be directly related to these hypotheses. Therefore, I will only review evidence that addresses the physical skill and creativity hypotheses and return to the social and training-for-the-unexpected hypotheses in the Discussion section.

Training Physical Skills

The beneficial effects of physical activity play on the development of physical skills in sports seems intuitive. Through physical activity play, athletes can acquire a multifaceted repertoire of perceptual-motor skills that facilitate the emergence of sport-specific functional skill solutions in later development (Pol et al., 2020; Schmidt & Wrisberg, 2008; Wolpert et al., 2011). The dynamic and complex nature of physical activity play has been suggested to provide an ideal situation for extensive implicit skill learning (Côté et al., 2007; Côté & Vierimaa, 2014; DiFiori et al., 2018; Maxwell et al., 2000; Memmert et al., 2015; Pol et al., 2020).

A common distinction in the skill acquisition literature in sports assumes two main categories of learning processes: explicit and implicit learning (Masters & Maxwell, 2004). Explicit learning is typically defined as verbally instructed, rule-based learning that depends on working memory to consciously encode declarative knowledge, whereas implicit learning is commonly described as a more automatic (procedural) learning process in which perceived patterns are encoded mostly nonconsciously. Implicit motor learning, therefore, is the acquisition of a new motor skill without a corresponding increase in verbal knowledge about the skill. Reber (1992) suggested that implicit learning is grounded in phylogenetically older structures and processes as explicit learning requires cognitive verbal structures that are assumed to be phylogenetically newer. As human ancestors were able to acquire skills before evolving the capacity for verbal communication, these skills must have been acquired by some nonverbal, implicit process.

Research has shown that implicit learning is an effective learning mechanism in golf (Masters, 1992), table tennis (Liao & Masters, 2001), whole-body balancing (Orrell et al., 2006; Shea et al., 2001), rugby (Poolton et al., 2007), basketball (Lam et al., 2009), and tennis (Farrow & Abernethy, 2002). Taken together these studies provide evidence for the evolutionary reasoning of Reber (1992) that motor skills learned implicitly may be more resilient under duress and less error prone as the phylogenetically older implicit learning system is better suited for perceptual-motor learning than the phylogenetically younger explicit learning system. Seger (1994) made a similar argument by stating that implicit and explicit learning may be specialized for certain situations with implicit learning playing a larger role in perceptual-motor learning or unstructured learning, whereas explicit learning may be used more in structured learning situations, such as formal schooling.

Several empirical studies in sports have indeed shown that verbal instructions from coaches can hinder motor learning or motor performance (e.g., Wulf & Lewthwaite, 2016 for a review). Verbal instructions have the potential to lead to "paralysis by analysis" in sport situations as they cause self-consciousness (Baumeister, 1984.; Hardy et al., 1996). This self-consciousness can cause athletes to turn their attention inward in an attempt to stabilize motor performance by explicitly monitoring skill execution. According to Duval and Wicklund (1972), this is because attention focused internally inevitably generates a self-evaluation process that controls whether the current standard of performance matches the standard of performance one has as one's goal (see Gallwey, 1997 for application of this in sports). Paradoxically, this has exactly the opposite effect than intended. Instead of facilitating sports performance various studies (e.g., Beilock et al., 2004; Gray, 2004) found evidence that this explicit attention disrupts well learned skills in golf or baseball, because the conscious system is too slow to deal with the real time control of motor skills. These studies provide empirical indication that an explicit, verbal approach to teaching physical skills can have detrimental effects and thereby provide a line of argumentation for an implicit approach to teaching sports by promoting free play. Afterall, motor skills like walking or running that humans are highly proficient at were never explicitly taught with the help of verbal instructions.

Fostering Creativity

> In acquiring skills, individuals (animals as well as humans) are in danger of finding suboptimal solutions to the many problems that confront them. In deliberately moving away from what might look like the final resting point, each individual may get somewhere that is better. Play may, therefore, fulfill an important probing role that enables the individual to escape a false endpoint. (Bateson, 2005, p. 17)

Similar to the reviewed examples of animals probing new behaviors during play (Burghardt, 2015) that later result in functional skills in the acquisition of food, there are many instances in which athletes created new techniques during play (e.g., the "Kempa Trick" in handball, the "Cruyff Turn" or the "Ronaldo Chop" in soccer). These sport-specific techniques resulted from play and have now become frequently used functional aspects of the respective games. Practice and play activities that promote exploration of movement patters enhance the variability of functional movement patterns in terms of both coordination and solutions. This has been shown to potentially lead to the emergence of novel, creative motor actions (Orth et al., 2017). Several lines of evidence in sports suggest that play can be an important means for developing domain-specific creativity and innovation (see Memmert, 2011 for a review).

Biographical data (Memmert et al., 2010) has suggested that physical activity play allows children the freedom to experiment with different movements and tactics, and, in turn, fosters domain-specific creativity and flexibility. Coaches from team sports (hockey, European handball, soccer, and basketball) were asked to select the most and least creative players from their teams. These players were then asked to provide information about the quantity and type of sport-specific practice and play activities undertaken throughout their careers. The most striking finding of the study was that the main difference between creative and noncreative players was the amount of time they had spent in unstructured play activities. More creative players had accumulated substantially more time in play activities than their less creative counterparts.

In further support of the importance of play in developing creativity in sports, Memmert and Roth (2007) conducted a longitudinal treatment study in which children participated in different training groups. All groups were tested with a sports creativity test before and after a 15-month treatment period. The results showed that diversified play led to significant improvements in measures of on-field creativity (standardized game test situations). Similar to the animal studies reviewed in the first section (i.e., playful interaction with objects led to novel adaptive behaviors in acquiring nutrition), these studies suggest that playful interaction in ball games facilitated the generation of functional new behaviors and enhanced problem-solving ability.

Research in sports has also shown that verbal instructions by coaches can have negative consequences in promoting learning and creativity in sports. A series of studies in the sports of European handball (Memmert & Furley, 2007) and basketball (Furley et al., 2010) demonstrated that verbal instructions from a coach intended to facilitate tactical decision making had the opposite effect and reduced the athletes focus of attention, leading them to miss unexpected open players. A control condition in the experimental paradigm verified that the players passed to these open players without the instructions from the coach. The authors argued that coaches frequently give verbal instructions as they consider this their job and feel incompetent if they instruct less (Furley et al., 2010). However, the results showed that this can reduce the domain-specific creativity of athletes by narrowing their focus of attention.

This reasoning received further support from a 6-month training study (Memmert, 2007) in which 6-year-old children participated in a training program twice a week. Two groups did the exact same games and exercises but received differed amounts of verbal instructions from coaches during the games. A third group acted as control group and only participated in regular physical education classes. All three groups were tested with a tactical creativity test before and after the 6-month period. The test involved a standardized game test situation in which the video-taped behavior of children was coded by trained experts. The results showed that the group receiving little verbal instructions from coaches significantly outperformed the other two groups in domain-specific creative performance gains.

Toward a Biocultural Framework for Positive Development and Success in Sports

It has recently been suggested that Darwin's theory of evolution by natural and sexual selection has the potential to add to the discussion of how to structure experiential activities in the acquisition of sport expertise (Winegard et al., 2018). In particular, an expansion of the behavioral repertoire increased the chances to cope with environmental variation and change (Geary, 2005). The most critical source of environmental variation in humans that would benefit from a broad flexible behavioral repertoire is the unpredictability of other people (Alexander, 1989). To anticipate other people's behavior and responding adaptively required humans to evolve means of acquiring cognitive and behavioral flexibility as "people with rigid, straightforward social algorithms would quickly be outmaneuvered" (Winegard et al., 2018, p. 41). It has also been suggested that ecological variability was a further important evolutionary driver of cognitive and behavioral flexibility (Potts, 2013). To achieve this flexibility, humans are believed to have acquired primary (evolved) and secondary (learned, culturally acquired) mechanisms of acquiring skills (Geary, 2005). Basic modular functions of the human brain are assumed to support universal primary means (e.g. rhythmic stereotypies like arm waving or foot kicking) that need to be subsequently "fine-tuned" and adapted to variable contexts through children's engagement in play and exploration. In later development this initial play behavior can be supplemented by coach-led practice in acquiring sport-specific expertise.

This biocultural theorizing is supported by research highlighting the benefits of first acquiring a broad range of skills through play activities in various sports before practicing specialized skills in one sport (Güllich, 2014, 2017; Güllich & Emrich, 2014; Hornig et al., 2016). In this respect, I consider George Kelly's (1955) metaphor of the creative cycle useful. According to this view the creative cycle involves two stages: loosening and tightening. These roughly correspond to random variation and selection stages in evolutionary theory. Loosening consists of the generation of novelty and should not be restricted as it is not clear which new variants will be successful. In the second stage, tightening, the exemplars that were generated during the loosening stage must be filtered out in a selection process to sort out those phenomena that have a chance to survive in competition. The empirical evidence in sports supports this loosening and tightening perspective by showing that early multisport play that subsequently transitions to domain-specific practice facilitates long-term development and success in sports (Fraser-Thomas et al., 2008a, 2008b; Güllich, 2017; Güllich & Emrich, 2014; Myer et al., 2015; Wall & Côté, 2007).

Beside the reviewed evidence on physical skill acquisition and fostered domain-specific creativity through play, several further arguments have been made why this loosening and tightening strategy might be beneficial in sports. Diversified play is likely to prevent the risk of overuse injury through less cumulative stereotypical mechanical impact on certain tissues (DiFiori et al., 2014; Hall et al., 2015; Jayanthi et al., 2015). It might also prevent early dropout and lead to prolonged participation (Butcher et al., 2002); Myer et al., 2015; Wall & Côté, 2007). Further, children that have explored various sports are likely to choses a primary sport independently that they enjoy and optimally suits them (Vaeyens et al., 2009). This autonomous choice is likely to increase intrinsic motivation and lead to more investment in the chosen primary sport later (Soberlak & Côté, 2003; Vallerand, 2001). Further, in terms of skill acquisition, early diversified play experience has been argued to lead to the acquisition of broad, transferable physical abilities, perceptual-motor, and psychological skills (Abernethy et al., 2005; Memmert & Roth, 2007; Rea & Lavallee, 2015). It has also been suggested to improve the efficiency of later practice in the primary sports (Güllich, 2014, 2017; Hornig et al., 2016; Rees et al., 2016). Finally, during playful activities children learn to adapt to new situations with more ease as they likely play in different roles/positions, vary rules, play on different surfaces, with different balls, court sizes, and with and against other children with varying skill levels (Davids et al., 2016; Memmert, 2011).

A Biocultural Perspective on Teaching in Sports

Other animals also acquire skills and information from others but experienced animals are not generally thought to actively teach (Danchin et al., 2004). From an evolutionary perspective, teaching can be defined as a "teacher" modifying his or her behavior in the presence of a "pupil" to facilitate the pupil's learning (Caro & Hauser, 1992). Although the frequency of animal teaching may have been underestimated (Hoppitt et al., 2008; Thornton & Raihani, 2008), the pervasiveness of human teaching offers a striking contrast to teaching in other animals (Fogarty et al., 2011). Mathematical models suggest that teaching is rare in nature as it would only evolve in very particular scenarios in which teaching would prove effective (Fogarty et al., 2011). Pertinent to the present line of argumentation, anthropological research has shown that verbal teaching mainly occurs in industrialized cultures (see Lancy, 2016 for a review) and is a rare means of facilitating learning among children in other human cultures (see also Gray, 2013). A recent anthropological study in Aka hunter-gatherers (Hewlett & Roulette, 2016) provided evidence for effective teaching of a variety of perceptual-motor skills described as natural pedagogy (e.g., demonstration, pointing, administering objects for experimentation). Importantly, this natural pedagogy has been shown to involve little explicit verbal instruction.

As many athletes, coaches, and parents believe that success in sports depends on hours of coach-led explicit teaching (DiFiori et al., 2018; Furley et al., 2010), there seems to be an overabundance of explicit verbal teaching in sports (Masters & Maxwell, 2004; Memmert, 2011). The reviewed evolutionary literature suggests that this might be counterproductive given children's evolved motivation to engage in natural playful activities. The outlined biocultural perspective on learning and teaching in sports suggests that a combination of early diversified play experiences (loosening) combined with subsequent coach led deliberate practice activities (tightening) is a promising route to a positive development and success in sports (Brenner, J. & Council on Sports Medicine & Fitness, 2016; Coté et al., 2009; Feeley et al., 2016; LaPrade et al., 2016; Myer et al., 2015; Rees et al., 2016) as it constitutes a better fit with human nature.

Discussion

The goal of the present article was to integrate evolutionary literature on physical activity play with literature from sports science to provide a coherent biocultural model of skill acquisition in sports. Most importantly, the present synthesis shows that the evolutionary theorizing on the functions of physical activity play proved helpful in explaining how diversified play experiences at an early age contribute to positive development and success in sports (e.g., Güllich, 2014, 2017; Güllich & Emrich, 2014; Hornig et al., 2016). The developmental scaffolding theory that the innate motivation of seeking pleasure in physical activity play coincidentally leads to the acquisition of physical skills and enhanced domain-specific creativity was supported by evidence from sports. A further important finding that emerged was that Western cultures often favor an explicit verbal approach of teaching specialized sports skills. However, this might collide with biologically prepared implicit learning and teaching mechanisms and lead to suboptimal development in sports.

The biocultural perspective on physical activity play in sports has the potential to inform and stimulate cross disciplinary research. For example, it has been argued that the ratio of theory to data is too high in evolutionary psychology (Ketelaar & Ellis, 2000), and I would argue that the ratio of theory to data is too low in sports. Although the situation has certainly improved in evolutionary psychology 20 years after Ketelaar and Ellis' (2000) observation (Buss, 2019 for a review), I am still convinced that both psychology and sport science can mutually benefit from using evolutionary inspired investigation on the growing body of sports data and documentation (Furley, 2019). The present review shows that sports data can be used to test evolutionary inspired hypothesis, for example on developmental scaffolding, practicing physical skills, and fostering domain-specific creativity. Further research might want to test the social training and training-for-the-unexpected hypothesis in sports. It seems feasible that some of the positive effects of early play experiences in sports on later success might be due to being socially competent (e.g., interacting better with teammates, adopting different roles in teams) and coping well with novel situations (e.g., responding to deceptive behavior of opponents, recovering from failure). In addition to testing such evolutionary inspired hypotheses using sports data, the field of sports science has recently been criticized for lacking a suitable overarching theoretical framework (Glazier, 2017) and scholars have critiqued the absence of evolutionary theorizing in the field of motor learning (Masters & Maxwell, 2004). Therefore, I consider evolutionary theory a suitable overarching framework for integrating a broad variety of findings in sports and deriving testable hypotheses on sports data.

Regarding the practical application of the biocultural framework, it is important to note that is not necessarily better or worse to be highly specialized or have a broad, diverse range of skills in sports. This depends on the context. A recent popular science book (Epstein, 2019) has outlined the early specialization and diversification pathways in acquiring expertise in domains like sports by describing the biographies of celebrity golfer Tiger Woods and tennis star Roger Federer. Tiger Woods serves as a prime example to illustrate the specialization pathway and Federer the diversification pathway. Evidently both achieved incredible success in their respective sports, although the biography of Tiger Woods was illustrative of some of the outlined problems that have been identified with the early specialization pathway in sports (Brenner, J. & Council on Sports Medicine & Fitness, 2016; Coté et al., 2009; Feeley et al., 2016; LaPrade et al., 2016; Myer et al., 2015; Rees et al., 2016). Also, different sports clearly require different combinations of practice and play activities and are likely to benefit more from diversification and others more from specialization. Broadly speaking, domains involving unpredictable patterns of human behavior (e.g., team ball games like soccer or basketball) in which patterns do not clearly repeat are likely to benefit more from diversified play activities than domains that are characterized by highly standardized and repetitive circumstances (e.g., track and field or darts). The fact that experiential learning activities are highly domain-specific has been repeatedly shown in various domains (e.g., Kahneman & Klein, 2009). Hence, narrow experience involving high-levels of repetitive deliberate practice is likely to be more effective in activities like darts or pole vaulting, but less effective in domains like soccer.

A helpful framework for structuring learning activities was proposed by Hogarth (2001). The framework specifies which learning activities work better in which contexts by distinguishing between learning (L) and performance settings (P). In some situations, the knowledge and skills acquired in L perfectly match the demands in P, whereas in other instances there is a less than perfect match between L and P, and again in others the knowledge and skills acquired in L might actually be misleading in P (Hogarth et al., 2015). Structuring L to meet the demands of P is more straightforward in some sports compared with others. In complex team-sports like soccer it is more difficult to create learning scenarios that exactly match the performance demands. Therefore, these domains are likely to benefit more from means of fostering variety and flexibility. Arguably, physical play activities are suitable in this endeavor and games like soccer therefore are likely to benefit from a higher ratio of play to practice (see Hornig et al., 2016), whereas domains like the 100-m sprint in which the challenges are highly similar and repetitive are likely to benefit more from a higher ratio of practice to play.

Although, a vast amount of literature emphasizes the importance of play in a child's physical, mental, social, and emotional development (Lewis, 2010; Piaget, 1962; Vygotsky, 1967), it remains hard to derive precise evidence-based recommendations on implementing physical activity play in organized sport and education programs. As a rule of thumb, adults (parents, coaches, and teachers) should promote diverse physical activity play without too much involvement of themselves at an early developmental stage. Physical activity play should constitute a safe environment in which a child is allowed to experiment, explore, and be creative without constant instruction and interruption from adults. In this respect, I consider the imperative “first do no harm” (which is part of the Hippocratic oath “Primum non nocere” that physicians have to take) a sensible guideline for adults. Evidence-informed developmental guidelines on implementing play activities are highly domain-specific (see e.g., diFiori et al., 2018 for the domain of basketball). Nevertheless, the reviewed evolutionary research (e.g., Pellegrini & Smith, 1998) indicates that sensitive periods in development exist (starting with the onset of locomotion at approximately 1 year) in which children would benefit most from spending substantial amounts of time in diverse physical play activities involving peers and variable environmental circumstances.

It is my hope that the present biocultural perspective on physical activity play leads to further cross-disciplinary research which might help to answer important basic and applied questions on physical activity play.

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